

CASE RECORDS of the MASSACHUSETTS GENERAL HOSPITAL

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Case 18-2026: A 53-Year-Old Man with Leg Weakness, Pain, and Weight Loss

Rabih Geha, M.D.,^{1,2} Nima Omid-Fard, M.D.,^{3,4}Michaël C.C. Slama, M.D., Ph.D.,^{5,6} and Haatem M. Reda, M.D.^{5,6}

PRESENTATION OF CASE

Author affiliations are listed at the end of the article.

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CME



Dr. Daniel H. Rubin (Medicine): A 53-year-old man presented to the emergency department of this hospital because of leg weakness, pain, and weight loss.

Five months before the current presentation, the patient began to have profound thirst, frequent urination, and fatigue. He established care with a local primary care clinic for evaluation of these symptoms. At that time, the patient also reported atraumatic pain in the left hip that radiated to the thigh. The pain was most prominent when he ascended stairs. He had had an 11-kg weight loss during the preceding year, which he had ascribed to a poor appetite. A physical examination, including neurologic examination, was reportedly normal. The glycated hemoglobin level was greater than 14.0% (reference range, 4.3 to 5.6). Results of additional laboratory testing, including blood levels of electrolytes, aspartate aminotransferase, alanine aminotransferase, albumin, and bilirubin, along with results of kidney-function tests and a complete blood count, were normal. Treatment with metformin and dulaglutide was initiated.

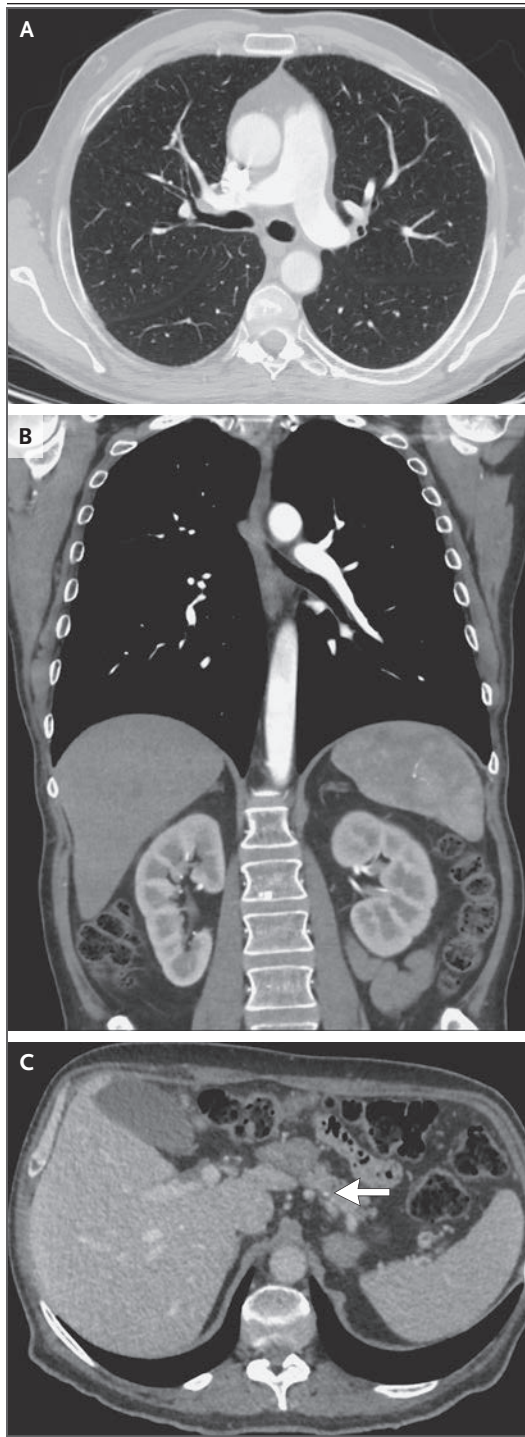
Four months before the current presentation, the patient had a telemedicine visit with his primary care provider. He reported progressive pain involving both hips and both thighs that was unresponsive to treatment with acetaminophen and ibuprofen, and he had begun using a cane for ambulation. The patient also had difficulty ascending the stairs in his house as a result of the pain. In addition, shivering and sweating occurred after administration of dulaglutide; these symptoms abated when the medication was withheld. Treatment with dulaglutide was discontinued, and insulin glargine was initiated.

Three months before the current presentation, the patient sustained a fall while trying to stand up from his bed. As a result, he sought care at the emergency department of this hospital. At that time, he noted new weakness and pain in both legs that worsened with ambulation and was accompanied by diminished sensation in both legs.

On examination, the temperature was 36.0°C, the heart rate 90 beats per minute, the blood pressure 126/68 mm Hg, the respiratory rate 16 breaths per minute, and the oxygen saturation 100% while the patient was breathing ambient air. Neuro-

Figure 1. CT Images of the Chest, Abdomen, and Pelvis.

Contrast-enhanced CT was performed 3 months before the current presentation. An axial image of the chest (Panel A) and a coronal image of the abdomen (Panel B) show no mass or other substantial abnormality; an image of the pelvis (not shown) also showed no such abnormalities. An axial image of the upper abdomen (Panel C) shows a tiny focus of hyperattenuation in the pancreas (arrow) that appears to be consistent with a vascular microcalcification. Otherwise, no mass or other pancreatic abnormality is seen.



logic examination revealed pain in the left leg during the straight-leg-raising maneuver and preserved distal strength in both legs; however, owing to pain, he was unable to lift the left leg against gravity. There was focal tenderness on palpation over the left greater trochanter and hip joint. Laboratory testing was notable for mild hyperglycemia, with a blood glucose level of 139 mg per deciliter (reference range, 70 to 99). The blood levels of thyrotropin, electrolytes, albumin, bilirubin, aspartate aminotransferase, and alanine aminotransferase were normal, as were the results of kidney-function tests.

Dr. Nima Omid-Fard: Computed tomography (CT) of the chest (Fig. 1A), abdomen (Fig. 1B), and pelvis (not shown), performed after the administration of intravenous contrast material as part of a trauma survey in the emergency department for recurrent unwitnessed falls, showed no spinal fracture, spinal malalignment, mass, lymphadenopathy, effusion in the hip joints, or radiologically significant degenerative change. On a CT image of the upper abdomen (Fig. 1C), a small focus of hyperattenuation was detected in the pancreas and appeared to be consistent with a vascular microcalcification. Otherwise, the pancreas appeared to be homogeneous and had a normal volume.

Dr. Rubin: Treatment with gabapentin was initiated. The patient declined to remain in the hospital while awaiting a physical therapy evaluation and was discharged home.

One month before the current presentation, the patient was seen for follow-up at his primary care clinic. He noted stable symptoms and continued to use assistive devices for ambulation. He was unable to work owing to incapacity and had transitioned to short-term disability. Regular laboratory evaluation showed that the glycated hemoglobin level had decreased to 10%. The dose

of gabapentin was increased, and the patient was referred for physical therapy and neurologic consultation.

On the day of the current presentation, the patient was sent from his outpatient physical therapy evaluation to the emergency department of this

hospital owing to clinically significant functional impairments. He was admitted directly to the medical service.

On admission, the patient noted progressive weakness, even with a walker, that caused sporadic buckling of the legs. He was no longer able to climb stairs in an upright position; instead, he had to push himself up each stair while in a

seated position. The patient noted an additional 18 kg of weight loss during the preceding 5 months. A review of systems was negative for night sweats, hematochezia, melena, diarrhea, rigors, rash, dyspnea, cough, saddle anesthesia, incontinence, urinary retention, headache, or bulbar symptoms. His medical history was notable for alcohol use disorder, which had been in re-

Table 1. Laboratory Data.*

Variable	Reference Range, Adults†	3 Mo before Presentation, Emergency Department, This Hospital	On Current Presentation, This Hospital
Hemoglobin (g/dl)	13.5–17.5	16.0	14.8
Hematocrit (%)	41.0–53.0	47.5	42.3
White-cell count (per μ l)	4000–11,000	7010	4400
Platelet count (per μ l)	150,000–400,000	240,000	273,000
Sodium (mmol/liter)	135–145	138	140
Potassium (mmol/liter)	3.4–5.0	4.2	3.5
Chloride (mmol/liter)	98–108	101	102
Carbon dioxide (mmol/liter)	20–31	24	25
Urea nitrogen (mg/dl)	6–23	9	10
Creatinine (mg/dl)	0.60–1.30	0.58	0.56
Calcium (mg/dl)	8.5–10.5	10.5	9.6
Glucose (mg/dl)	70–99	139	120
Aspartate aminotransferase (U/liter)	10–40	11	14
Alanine aminotransferase (U/liter)	10–55	8	7
Alkaline phosphatase (U/liter)	45–115	99	57
Total bilirubin (mg/dl)	0.0–1.2	0.5	0.6
Albumin (g/dl)	3.3–5.2	4.8	4.3
Globulin (g/dl)	1.9–4.1	2.3	2.2
Vitamin B ₁₂ (pg/ml)	232–1245	—	662
Folate (ng/ml)	>4.7	—	12.5
Creatine kinase (U/liter)	39–308	—	47
C-reactive protein (mg/liter)	<10.0	—	0.2
Erythrocyte sedimentation rate (mm/hr)	0–20	—	9
IgG (mg/dl)	700–1600	—	466
IgA (mg/dl)	70–400	—	253
IgM (mg/dl)	40–230	—	36
Free kappa light chain (mg/liter)	3.3–19.4	—	11.5
Free lambda light chain (mg/liter)	5.7–26.3	—	15.7
Ratio of kappa to lambda free light chains	0.26–1.65	—	0.73
Serum protein electrophoresis	Negative for monoclonal component	—	Negative for monoclonal component

* To convert the values for urea nitrogen to millimoles per liter, multiply by 0.357. To convert the values for creatinine to micromoles per liter, multiply by 88.4. To convert the values for calcium to millimoles per liter, multiply by 0.250. To convert the values for glucose to millimoles per liter, multiply by 0.05551. To convert the values for bilirubin to micromoles per liter, multiply by 17.1. To convert the values for vitamin B₁₂ to picomoles per liter, multiply by 0.7378. To convert the values for folate to nanomoles per liter, multiply by 2.266.

† Reference values are affected by many variables, including the patient population and the laboratory methods used. The ranges used at Massachusetts General Hospital are for adults who are not pregnant and do not have medical conditions that could affect the results. They may therefore not be appropriate for all patients.

mission for more than a decade, as well as generalized anxiety disorder. He had not had a colonoscopy as part of age-appropriate cancer screening. Medications included sertraline, gabapentin, glargine, and metformin. The patient resided alone in a suburban area of New England and worked as an administrative assistant.

He did not use recreational drugs but had smoked one pack of cigarettes per day for 35 years. The patient had sex with men; he had had no sexual partners in the preceding 2 years and had no history of sexually transmitted infections.

On examination, the temporal temperature was 36.7°C, the heart rate 94 beats per minute, the

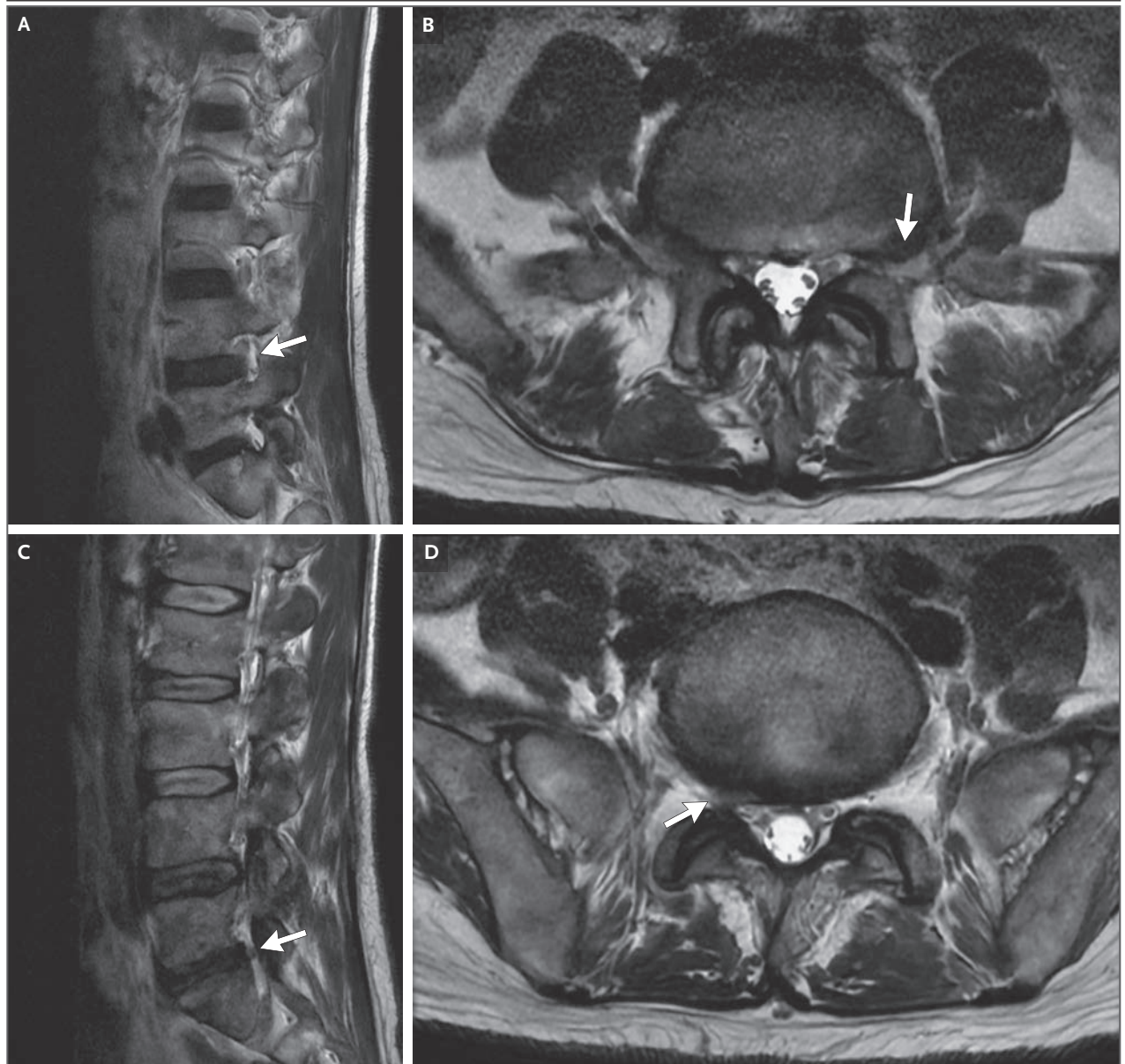


Figure 2. MRI of the Lumbar Spine.

MRI of the lumbar spine was performed at the time of the current presentation. Sagittal (Panel A) and axial (Panel B) T2-weighted images show a small extraforaminal disk protrusion at the L4–L5 level on the left side (arrows) that is causing minimal contact with the exiting left L4 nerve root, without radiologically significant foraminal stenosis. Sagittal (Panel C) and axial (Panel D) T2-weighted images obtained at a different level show a small subarticular disk protrusion at the L5–S1 level on the right side (arrows) that is causing minimal contact with the descending right S1 nerve root, without radiologically significant spinal-canal stenosis.

blood pressure 128/87 mm Hg, the respiratory rate 16 breaths per minute, and the oxygen saturation 100% while the patient was breathing ambient air. The weight was 73.5 kg, and the body-mass index (the weight in kilograms divided by the square of the height in meters) was 20.8. He appeared tired. No abnormalities were noted on cardiopulmonary auscultation, and the abdomen was soft and nontender. The arms and legs were warm and without edema. Neurologic examination showed symmetric cranial-nerve function with preserved strength and normal sensation in the arms. Strength on hip flexion and knee extension was diminished to 2/5 on the left side and to 3/5 on the right side. Distal leg strength was preserved in the feet, and a mild loss of sensation to touch was noted in the toes. The reflexes in the legs were absent. Asymmetric muscle atrophy was seen in the thighs, with the left leg affected to a greater extent than the right leg; sporadic fasciculations were noted in the left thigh muscle.

Blood levels of vitamin B₁₂, folate, creatine kinase, and C-reactive protein were normal, as was the erythrocyte sedimentation rate. Tests for human immunodeficiency virus, *Treponema pallidum*, and *Borrelia burgdorferi* were negative. Serum protein electrophoresis showed no monoclonal protein, and a serum free light-chain assay showed a normal light-chain ratio (ratio of kappa to lambda light chains); quantitative serum IgG and IgM levels were 466 mg per deciliter (reference range, 700 to 1600) and 36 mg per deciliter (reference range, 40 to 230), respectively. Additional laboratory test results are shown in Table 1.

Dr. Omid-Fard: Magnetic resonance imaging (MRI) of the thoracic and lumbar spine, performed without the administration of intravenous contrast material, showed mild spondylosis, with a small extraforaminal disk protrusion at the L4–L5 level on the left side that caused minimal contact with the left L4 nerve root (Fig. 2A and 2B) and a subarticular disk protrusion at the L5–S1 level on the right side that caused minimal contact with the right S1 nerve root (Fig. 2C and 2D), without radiologically significant neural foraminal or spinal-canal stenosis. No thoracolumbar mass, cord signal abnormality, nerve-root thickening, or epidural collection was observed. There was no paraspinal edema or facet joint effusion.

Dr. Rubin: A diagnostic test was performed.

DIFFERENTIAL DIAGNOSIS

Dr. Rabiha Geha: This 53-year-old man presented with a subacute syndrome characterized by progressive, asymmetric weakness and marked weight loss against a backdrop of newly diagnosed diabetes. Beyond the patient's notable neurologic abnormalities and hyperglycemia, the clinical laboratory and imaging evaluations were strikingly unrevealing, showing only hypogammaglobulinemia.

HYPOGAMMAGLOBULINEMIA

The patient had hypogammaglobulinemia that was characterized by low circulating levels of both IgG and IgM. Hypogammaglobulinemia can arise from a variety of mechanisms. In adults, it is often caused by a primary immunodeficiency syndrome such as common variable immune deficiency, medications such as glucocorticoids, or abnormalities that lead to protein loss, such as nephrotic syndrome or protein-losing enteropathies. This patient's presentation lacked features associated with these conditions, and therefore, additional explanation should be sought. Although uncontrolled diabetes mellitus has been implicated as a cause of hypogammaglobulinemia,¹ an underlying lymphoproliferative disorder would explain both the low immunoglobulin levels and the weight loss, which makes such a disorder an appealing unifying diagnosis. However, the absence of lymphadenopathy, splenomegaly, and cytopenias in this patient is not consistent with a lymphoproliferative disorder such as an aggressive lymphoma.

DIABETES MELLITUS

Type 2 diabetes mellitus typically occurs as a complex interplay of genetic and metabolic factors that culminate in insulin resistance and hyperglycemia. However, a minority of cases arise from a single cause, including treatment with glucocorticoids or atypical antipsychotic medications, cancer, pancreatic insufficiency, or an endocrinopathy such as hypercortisolism. The combination of new-onset, severe diabetes mellitus accompanied by profound weight loss prompts consideration of pancreatic cancer or insulin deficiency. Although a small pancreatic neuroendocrine tumor may be missed without dedicated pancreatic imaging such as endoscopic ultrasonography, the patient's pancreas appeared morphologically normal, with-

out masses or atrophy, which makes a primary pancreatic cause unlikely.²

Despite weight loss and a markedly elevated blood glucose level, the patient does not appear to be in a state of low insulin, as evidenced by the absence of ketosis and anion-gap metabolic acidosis. A small proportion of patients with type 2 diabetes mellitus may present with weight loss and symptoms associated with a hyperosmolar state³; their severe diabetes mellitus could be a dominant driver of the presentation despite preserved insulin production.

NEUROLOGIC SYNDROME

Perhaps the most striking features of this patient's problem representation are his pain and weakness. The dominant neurologic syndrome is characterized by pain and subacute, sequential, proximal weakness that is confined to the legs. The mild distal sensory loss may reflect concurrent diabetic peripheral neuropathy. The absence of reflexes in the legs further localizes the lesion to the peripheral nervous system, and the presence of fasciculations supports a neurogenic process. The degree of pain and the proximal predominance of the findings would be atypical for peripheral neuropathy alone; the neuroanatomical localization points to the nerve roots or plexus. The absence of bowel or bladder dysfunction favors a disorder of the nerve bundle (plexopathy) rather than a polyradiculopathy (affecting multiple spinal nerve roots).

PLEXOPATHY

Plexopathies typically result from compression, radiation, or trauma — all of which have been ruled out in this patient. Nerve plexuses can also be affected by infiltrative cancers, vasculitides, and neurotropic viral infections. It is important to note that infiltrative cancers may produce a plexopathy that can be difficult to detect; identification depends largely on performing MRI, particularly with the use of intravenous contrast material. Therefore, an infiltrative cancer has not been ruled out entirely as the cause of this patient's syndrome.

WEIGHT LOSS

The patient's loss of nearly 30 kg of weight further heightens concern about an occult cancer

that could be driving his presentation; however, cross-sectional imaging has ruled out widespread metastatic disease from solid tumors or hematologic cancers. Localized tumors rarely produce marked weight loss unless they impair nutritional intake or absorption directly. These tumors, which involve the upper gastrointestinal tract, often feature prominent symptoms — dysphagia, abdominal pain, or early satiety — all of which are absent in this patient. Furthermore, the patient's normal levels of inflammatory markers argue against cancer-related cachexia.⁴

In contrast, the catabolic state that can occur in patients with severe diabetes mellitus can be associated with minimal systemic inflammation. It is notable that diabetes mellitus is also uniquely associated with lumbosacral plexopathy, which often extends to the nerve roots and peripheral nerves — a condition known as diabetic lumbosacral radiculoplexus neuropathy (DLRPN).⁵ Imaging findings in persons with DLRPN may be completely normal.

DIABETIC LUMBOSACRAL RADICULOPLEXUS NEUROPATHY

DLRPN, also known as diabetic amyotrophy, is a rare complication of diabetes mellitus that is often diagnosed at the time of, or shortly after, diagnosis of type 2 diabetes mellitus. This patient's clinical course is typical of the syndrome, which begins with painful, predominantly proximal weakness in one leg and then spreads to the distal legs during the ensuing months. The severity of pain and weakness results in substantial disability, as was seen in this patient. In parallel, marked weight loss occurs, which leaves the patient in visible decline with pain, weakness, and muscle wasting. Paradoxically, notable laboratory and imaging findings are often absent. This discrepancy is not a contradiction but rather a signature of the disease and helps to rule out alternative diagnoses. Electrodiagnostic studies are necessary to provide further diagnostic support for this diagnosis, and therefore, I would recommend performing electromyography and nerve conduction studies.

DR. RABIH GEHA'S DIAGNOSIS

Diabetic lumbosacral radiculoplexus neuropathy.

A Sensory Nerve Conduction Studies

Nerve	Site	Distance <i>cm</i>	Peak Latency <i>msec</i>	Amplitude μ V	Temperature $^{\circ}$ C
Left ulnar	Fifth digit	11	No response	No response	30.6
Left radial–snuff box	Forearm	10	2.7	10.9	33.6
Right sural–lateral malleolus	Calf	14	4.7	4.5	32.4
Left sural–lateral malleolus	Calf	14	No response	No response	29.3
Right superficial peroneal	Lateral leg	14	No response	No response	31.7
Left superficial peroneal	Lateral leg	14	No response	No response	31.3

Mildly prolonged peak latencies could suggest mild myelin dysfunction (common in patients with diabetes), although process is predominantly axonal

All sensory response amplitudes are reduced to varying degrees $\rightarrow$ generalized or multifocal, postganglionic, axonal neurogenic process, with a non–length-dependent pattern

Absent ulnar response could be result of unrelated ulnar mono-neuropathy (common in patients with diabetes)

Marked asymmetry in the legs

B Motor Nerve Conduction Studies

Nerve	Site	Distance <i>cm</i>	Latency <i>msec</i>	Amplitude <i>mV</i>	Velocity <i>m/sec</i>	Temperature $^{\circ}$ C
Left ulnar–ADM	Wrist	6	2.8	7.4		33.5
	Below elbow	25	7.7	5.9	50.4	33.5
	Above elbow	10	10.0	5.8	44.9	33.5
Right peroneal–EDB	Ankle	9	5.5	0.9		
Left peroneal–EDB	Ankle	9	5.5	0.2		32.1
	Below fibular head	43	14.4	0.1	48.3	32.1
	Above fibular head	13	17.7	0.1	39.2	31.9
Right tibial–AH	Ankle	10	5.4	0.5		32.7
Left tibial–AH	Ankle	10	6.0	0.4		31.7
	Popliteal fossa	47	20.8	0.4	31.8	31.6

Low motor response amplitudes with essentially unaffected distal latencies and conduction velocities suggest predominantly axonal process

Asymmetry in the legs but both clearly affected
Low-amplitude right tibial motor response despite relatively preserved right sural sensory response suggests preganglionic component (e.g., radiculopathy)
Low-amplitude left tibial and bilateral peroneal motor responses, together with absent corresponding sensory responses, suggest postganglionic component (e.g., lumbosacral plexopathy vs. polyneuropathy vs. mono-neuropathies)

Mild slowing of conduction velocity suggests mild myelin dysfunction (from diabetes vs. ulnar neuropathy at the elbow)

C Needle Electromyography

Muscle	Spontaneous			Motor-Unit Action Potential			Recruitment		
	Fib or PSW	Fasc	Misc	Duration	Amplitude	Polyphasic	No. of MUs	Rate	Effort
Right deltoid	None	None	None	Normal	Normal	Normal	Normal	Normal	Full
Right first dorsal interosseous muscle	None	None	None	Normal	Normal	Normal	Normal	Normal	Full
Right vastus medialis	2+	None	None	Normal	Normal	Normal	Severe reduction	Rapid	Full
Right tibialis anterior	None	None	None	Long	High	Normal	Mild reduction	Rapid	Full
Right gastrocnemius muscle	3+	None	None	—	—	—	None	—	Submax

Presence of Fibs or PSWs suggests active denervation or incomplete reinnervation, process longer than 4–6 wk
Abnormalities despite relatively spared right sural sensory response suggest preganglionic component

Normal MUAP morphologic features despite severely reduced recruitment in right vastus medialis suggest duration of less than 6 mo

Mild chronic reinnervation changes in right tibialis anterior muscle may be unrelated to current presentation

Reduced recruitment pattern confirms neurogenic process

Figure 3 (facing page). Nerve Conduction Studies and Electromyography.

The sensory nerve conduction studies (Panel A) show reduced amplitudes in multiple nerves, which suggests a generalized or multifocal, asymmetric, predominantly axonal, postganglionic neurogenic process that is more severe in the left leg than in the right leg. The motor nerve conduction studies (Panel B) confirm an asymmetric, predominantly axonal, neurogenic process in both legs but also show a pattern that suggests a superimposed preganglionic process in the right leg. Needle electromyography (EMG) of the right leg (Panel C) shows a relatively severe subacute neurogenic process with incomplete reinnervation. The overall electrophysiological findings are consistent with a severe, subacute, asymmetric bilateral lumbosacral radiculoplexus neuropathy with incomplete reinnervation. A mild superimposed generalized, axonal sensory polyneuropathy and a mild nonlocalizing, left ulnar sensory mononeuropathy are also present; these findings were identified incidentally. ADM denotes abductor digiti minimi, AH abductor hallucis, EDB extensor digitorum brevis, Fasc fasciculation potential, Fib fibrillation potential, Misc miscellaneous, MU motor unit, MUAP motor-unit action potential, PSW positive sharp wave, and Submax submaximal.

DIAGNOSTIC TESTING

Dr. Michaël C.C. Slama: Electromyography and nerve conduction studies were performed. Before describing the findings in this patient, it is useful to provide an overview of the way that electrodiagnostic studies can help localize lesions involving the peripheral nervous system and muscles. Nerve conduction studies are usually performed first. In these studies, electrical stimuli are applied to sensory and motor nerves. Electrical responses are recorded, and measurements of amplitude, latency, and conduction velocity are derived. After the nerve conduction studies have been completed, electromyography is performed. Electromyography involves the insertion of a small concentric needle electrode into selected muscles. Electrical activity is assessed when the patient is at rest and during voluntary muscle contraction. The overall pattern of abnormalities allows precise localization of a lesion involving the peripheral nervous system and can provide additional information regarding pathophysiological mechanisms (e.g., an axonal process vs. a demyelinating process), severity, and chronicity.

In this patient, the sensory nerve conduction studies (Fig. 3A) showed reduced amplitudes in

multiple nerves, which suggest a generalized or multifocal, asymmetric, predominantly axonal, postganglionic neurogenic process such as polyneuropathy, plexopathy, or multiple mononeuropathies. The findings were more severe in the left leg than in the right leg. The motor nerve conduction studies (Fig. 3B) confirmed an asymmetric, predominantly axonal, neurogenic process in both legs; however, the pattern of abnormalities in the right leg also suggests a preganglionic component (i.e., lumbosacral radiculopathy).

Needle electromyography of the right leg (Fig. 3C) confirmed the presence of a neurogenic process on the basis of the reduced recruitment pattern of motor-unit action potentials (MUAPs). The abnormal spontaneous activity with frequent fibrillation potentials and positive sharp waves in several muscles suggest a process that had been present longer than 4 to 6 weeks with active denervation or incomplete reinnervation. The normal MUAP morphologic features without remodeling in the right vastus medialis muscle suggest a subacute time course of less than 6 months.

The relatively preserved right sural sensory response seen on nerve conduction studies, together with the markedly abnormal neurogenic findings in the right leg seen on electromyography, further supports the presence of a preganglionic component. The basis for this conclusion relates to the preganglionic pathological process, in which Wallerian degeneration of the sensory axons does not occur distal to the dorsal root ganglia, where the sensory nerve conduction studies are measured. The degree of spontaneous activity and recruitment abnormalities suggest a severe process.

Overall, the electrophysiological findings are most consistent with a severe, subacute, asymmetric bilateral lumbosacral radiculoplexus neuropathy with incomplete reinnervation. This pattern is commonly seen in patients with DLRPN.

Other electrophysiological abnormalities that were identified in this patient were incidental and included a mild superimposed generalized, non-length-dependent sensory or sensorimotor polyneuropathy with predominantly axonal pathophysiological characteristics and with features of mild myelin dysfunction. These findings are most likely related to the patient's diabetes mellitus, although the non-length-dependent pat-

tern would be atypical for diabetic peripheral neuropathy. An additional incidental finding was a mild left ulnar sensory mononeuropathy that did not clearly localize to the elbow region.

ELECTROPHYSIOLOGICAL DIAGNOSIS

Diabetic lumbosacral radiculoplexus neuropathy.

DISCUSSION OF MANAGEMENT

Dr. Haatem M. Reda: Vasculitic ischemia and neuritis have been among the postulated causes of DLRPN since the disease was first described in 1890.⁶ However, it was not until the late 1990s that pathological studies of sural and superficial fibular nerves in patients with DLRPN identified the presence of microscopic vasculitis of the vasa nervorum, a finding that suggested a potential benefit with immunotherapy.⁶

In a retrospective series involving patients with subacute diabetic proximal neuropathy who received treatment with glucocorticoids, intravenous immune globulin, or plasma exchange, 75% of the patients had at least some motor recovery. However, 59% of the untreated patients also had improvement, which highlights the probable self-limited and monophasic nature of this disorder regardless of glycemic control.⁷ Larger studies, including a randomized, controlled trial, have failed to show improved motor outcomes with immunomodulatory therapy; however, it appears that glucocorticoid therapy may reduce the severity and duration of pain.⁸

Once DLRPN is diagnosed, management is primarily supportive. Perhaps surprisingly, glycemic control does not appear to be a disease-modifying factor, and in fact, overly aggressive glycemic control may even provoke DLRPN.⁹ The evidence for physical therapy in patients with DLRPN is observational and is extrapolated from general rehabilitation principles. Starting physical and occupational therapy early appears to be beneficial. Gait retraining often involves at least temporary use of assistive devices such as ankle-foot orthoses, canes, and walkers.

Although weakness abates in almost all patients, most do not recover completely. Approximately 50% of patients use a wheelchair at some point during their disease course, and approximately 9 to 16% of patients ultimately use a

wheelchair indefinitely. Recovery takes at least several months and often up to a few years. Between 80 and 94% of patients report at least some residual motor deficits, most often foot drop on one or both sides.¹⁰

For pain management, much of the evidence is extrapolated from studies in diabetic neuropathic pain, as well as from anecdotal experience and expert consensus. Treatment strategies are the same for DLRPN as for other chronic neuropathic pain syndromes and include medications such as gabapentinoid agents, tricyclic antidepressants, and duloxetine as first-line therapeutic options. In up to 40% of patients, these medications show equivalent efficacy with at least 50% pain relief, as measured by scores on visual analogue scales. Second-line therapeutic options include sodium-channel blockers such as carbamazepine and valproic acid, other anticonvulsant medications such as lacosamide, or alpha lipoic acid. More invasive options, such as nerve blocks, may be considered in consultation with a pain medicine specialist.

Patients with DLRPN who receive glucocorticoid therapy have substantially less pain during the initial 6 months after diagnosis than untreated patients.⁸ Although glucocorticoids are not routinely recommended, weekly doses of intravenous methylprednisolone can be considered in patients with severe pain that is refractory to usual treatment or in those with relentless or progressive weakness. However, this strategy poses additional challenges for achieving adequate glycemic control in patients with diabetes mellitus. In a prospective open-label trial of intravenous immune globulin, four of five patients had a rapid reduction in pain within 2 weeks after treatment, but confirmation of these results in larger studies is lacking.¹¹ Intravenous immune globulin is an appealing immunomodulator, given that its glycemic profile is more favorable than that of glucocorticoids in patients with diabetes mellitus, so it may be considered for off-label use in selected cases. As with other causes of chronic neuropathic pain, treatment with full opioid agonists should be avoided, if possible.

FOLLOW-UP

Dr. Rubin: The patient was discharged to an acute rehabilitation hospital on hospital day 9. Treatment with pregabalin was added to his current

regimen with gabapentin. Treatment with sertraline was transitioned to duloxetine, which led to a decrease in pain. The patient remained in the hospital, where he underwent rehabilitation for an additional 3 weeks before being discharged home. At the time of discharge, he was ambulating with a walker independently but was also using a wheelchair owing to fatigue.

FINAL DIAGNOSIS

Diabetic lumbosacral radiculoplexus neuropathy.

CASE RECORDS EDITORS' NOTE — LESSONS LEARNED

1. A rare complication of diabetes mellitus is lumbosacral plexopathy, which often extends to the nerve roots and peripheral nerves — a condition known as diabetic lumbosacral radiculoplexus neuropathy (DLRPN, also known as diabetic amyotrophy).

2. Diagnosis of DLRPN often occurs at the

time of, or shortly after, diagnosis of type 2 diabetes mellitus.

3. The typical clinical syndrome begins with painful, predominantly proximal weakness in one leg and then spreads to the distal legs during the ensuing months. Profound weight loss may be a prominent clinical feature.

4. Imaging findings in persons with DLRPN may be normal, and electromyography and nerve conduction studies can help support the diagnosis.

5. DLRPN is typically a self-limited complication of diabetes mellitus, and management is primarily supportive. Glycemic control does not appear to be a disease-modifying factor.

This case was presented at the Medicine Case Conference.

Disclosure forms provided by the authors are available with the full text of this article at NEJM.org.

AUTHOR INFORMATION

¹Department of Medicine, University of California, San Francisco, San Francisco; ²Department of Medicine, San Francisco VA Medical Center, San Francisco; ³Department of Radiology, Massachusetts General Hospital, Boston; ⁴Department of Radiology, Harvard Medical School, Boston; ⁵Department of Neurology, Massachusetts General Hospital, Boston; ⁶Department of Neurology, Harvard Medical School, Boston.

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